FARAH MKAOUAR

End-of-studies internship – Data science

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PROFILE

Third-year engineering student in statistics at ESSAI, Tunis. I am comfortable with statistics, machine learning and generative AI, and I have applied them to banking, insurance, healthcare and retail. I am looking for an end-of-studies internship in data science, AI or risk modelling, in Tunisia or abroad.

EDUCATION

Engineering degree in statistics and data analysis | ESSAI, University of Carthage, Tunis | 2024 – 2027

Preparatory classes in mathematics and physics | IPEIS, Sfax | 2022 – 2024

Baccalaureate in mathematics, highest honours | Lycée Monji Slim, Sfax | 2022

PROFESSIONAL EXPERIENCE

Banking risk and data intern | UBCI, Tunis | 2026

Built an application to backtest internal rating models, tested on simulated data. Contributed to the bank's internal chatbot.

Data intern | Emerging Diseases Observatory | 2026

Built a hepatitis C screening application with patient anonymisation and a monitoring dashboard.

Quality and web development intern | SIDILEC International | June – July 2025

Quality diagnosis (Ishikawa, Pareto) and a web app for sampling inspection under ISO 2859-1 (React, PostgreSQL).

PROJECTS

- **AI agent for credit pre-qualification, team of two**: LLM agent that looks up customer data, simulates the loan and searches the credit policy (RAG), with personal-data masking and hand-off to an advisor. Tested on 1,287 conversations: accuracy from 44% to 95%, numerical errors from 82% to 4%.
- **IFRS 9 and stress testing**: estimated PD, LGD and EAD on 2.26 million loans, computed expected credit losses by stage and ran macroeconomic stress scenarios. Python, Streamlit.
- **ActuaNLP, team of two**: automated reading of 48 PDF annual reports (OCR, page classification with an F1 of 0.90, table extraction, question answering).
- **Airbnb price model**: model on 485,000 listings in 15 cities, including review text; R^2 of 0.815 and well-calibrated prediction intervals. LightGBM, PyTorch.
- **Motor insurance pricing**: large-claim threshold with the POT method on 21,600 contracts, then claim prediction on 58,600 policies. Python.
- **Multi-agent claims processing (in progress)**: extraction, coverage-check, fraud and settlement agents coordinated with LangGraph, compared with a single agent.
- **Credit scoring and internal rating**: logistic regression, Information Value variable selection, validation (AUC 0.82, Gini 0.64, KS 0.59) and a 4-grade rating scale. Python, R.
- **Deep hedging (in progress)**: PPO agent hedging an option in a stochastic-volatility (Heston) environment, compared with delta hedging.
- **E-commerce products**: scraped 290 products, segmentation (K-Means, PCA), price prediction from text and Power BI dashboards.

SKILLS

Programming: Python, R, SAS, SQL

Statistics and machine learning: regression, classification, scikit-learn, XGBoost, LightGBM, PyTorch, SHAP, statistical testing

Generative AI: LLM agents, RAG, OCR, NLP

Data and tools: Power BI, Excel, Streamlit, FastAPI, React, PostgreSQL, Docker, Git

Domains: credit risk (IFRS 9, scoring), non-life actuarial work, healthcare, retail

ACTIVITIES

Kaggle (Go Data Science 5.0) | 3rd prize, Green Lab Hackathon | Enactus ESSAI | ML Club ESSAI

LANGUAGES

Arabic (native) | French (fluent, DELF 2026) | English (B2)